

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8391**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	305	Seat No:	2022700057
Course Section:	1				

L_41_670_986_L_1505_41387



Name of the candidate. Deep Solanki

Candidate's UID number:

2	0	2	2	7	0	0	0	5	7
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Examination class room number:

3	0	5
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Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

PROGRAM: Data Science

DAY AND DATE: 8th October, 2025

COURSE NAME: Deep Learning

SEMESTER :

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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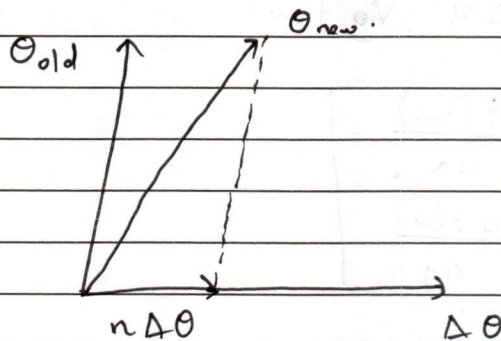
Q1) a) Let us consider a vector of parameters.

$$\theta = [w \ b]$$

$$\therefore \Delta\theta = [\Delta w \ \Delta b]$$

Let n be a very small value.

$$\theta_{\text{new}} = \theta_{\text{old}} + n \Delta\theta$$



$\Delta\theta$ can be obtained from Taylor series;

$$L(\theta_{\text{new}}) = L(\theta_{\text{old}}) + n u^T \nabla_{\theta} L(\theta) + \frac{n^2}{2!} u^T \nabla_{\theta}^2 L(\theta) u$$

+ ...

Since n is very small, ignore n^2 & higher degree n terms.

$$L(\theta_{\text{new}}) = L(\theta_{\text{old}}) + n u^T \nabla_{\theta} L(\theta)$$

$$L(\theta_{\text{new}}) - L(\theta_{\text{old}}) < 0$$

$$\therefore u^T \cdot \underbrace{\nabla_{\theta} L(\theta)}_{\substack{\text{Gradient} \\ \text{Dot prod.}}} < 0$$

$$-1 \leq \cos \beta = \frac{u^T \times \nabla_{\theta} L(\theta)}{|u^T| \times |\nabla_{\theta} L(\theta)|} \leq 1$$

$$k = |u^T| \times |\nabla_{\theta} L(\theta)|$$

Multiply by k through out;

$$-k \leq \cos \beta = u^T \times \nabla_{\theta} L(\theta) \leq k$$

↓

$$k = -1$$

$$\cos \beta = -1$$

$$\therefore \beta = 180^\circ$$

$\therefore u$ should move in direction opposite to the gradient.

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$$f(w) = \frac{1}{2} |Xw - y|^2$$

$$= \frac{1}{2} |f(x) - y|^2$$

$$f(x) = Xw$$

$$\therefore \text{Change} = n u^T \nabla_w f(w)$$

$$\text{Gradient} = \begin{bmatrix} \frac{df(w)}{dx} \\ \frac{df(w)}{dy} \end{bmatrix}$$

$$\frac{df(w)}{dx} = \frac{1}{2} \times (2 |Xw - y| \times \frac{d(Xw - y)}{dx})$$

$$= \frac{1}{2} \times (2 |Xw - y| \times w)$$

$$= (Xw - y)w$$

$$= w \cdot f(w)$$

$$\frac{df(w)}{dy} = \frac{1}{2} \times 2 |Xw - y| \times \frac{d(Xw - y)}{dy}$$

$$= |Xw - y| \times -1$$

$$= -f(w)$$

$$\text{Gradient} = \begin{bmatrix} w \cdot f(w) \\ -f(w) \end{bmatrix}$$

$$\nabla_w^2 f(w) = \begin{bmatrix} \frac{d^2 f(w)}{dx^2} & \frac{d^2 f(w)}{dx dy} \\ \frac{d^2 f(w)}{dy dx} & \frac{d^2 f(w)}{dy^2} \end{bmatrix}$$

$$= \begin{bmatrix} w \times w \times f(w) & -w \cdot f(w) \\ -w \cdot f(w) & f(w) \end{bmatrix}$$

$$= \begin{bmatrix} w^2 f(w) & -w \cdot f(w) \\ -w \cdot f(w) & f(w) \end{bmatrix}$$

Hessian matrix



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$$\begin{vmatrix} \omega^2 \cdot f(\omega) - \lambda & -\omega \cdot f(\omega) \\ -\omega \cdot f(\omega) & f(\omega) - \lambda \end{vmatrix} = 0$$

$$(\omega^2 \cdot f(\omega) - \lambda)(f(\omega) - \lambda) - \omega^2 f^2(\omega) = 0$$

$$\omega^2 \cdot f^2(\omega) - \lambda \cdot \omega^2 f(\omega) - \lambda f(\omega) + \lambda^2 - \omega^2 f^2(\omega) = 0$$

$$\lambda^2 - \lambda(\omega^2 f(\omega) + f(\omega)) = 0$$

$$\lambda - \omega^2 f(\omega) + f(\omega) = 0$$

$$\lambda = \omega^2 f(\omega) + f(\omega), \quad 0$$

↓

∴

$$\therefore 0 < n < \frac{2}{\lambda_{\min}}$$

$$0 < n < \frac{2}{\omega^2 \cdot f(\omega) + f(\omega)}$$

λ_{min}

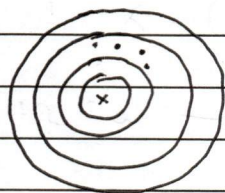
ω² · f(ω) + f(ω)

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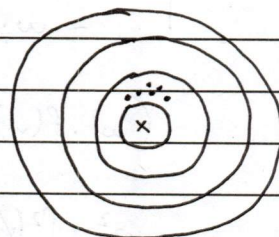
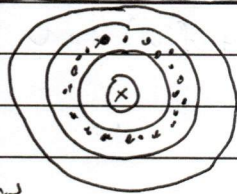
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variance $\rightarrow$ spread
bias $\rightarrow$ distance from
the target

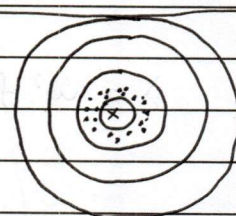
Q. 1b) Bias variance tradeoff :-

High
~~Low~~ bias

Low variance

Low
~~High~~ bias~~Low~~ varianceHigh
~~Low~~ bias

High variance

Low bias
~~High bias~~

High variance

- The model with low bias & high variance is a complex model. There is a problem of overfitting.

$\rightarrow$ Overfitting occurs when the model learns from the train data (even the noise) & thus performs poorly on the test data. Hence, its accuracy during training is much more than testing.

It also affects generalization as there is too much difference in training & testing.

Hence we can reduce overfitting by:-

- L1/L2 regularizations - We can assign weight to certain data points which are noise & minimize or remove them in order to obtain a more generalized model.
- Rather than providing a lot of data, we can divide the data into batches & then train these batches independently. Thus, the probability of learning noise from the data decreases.

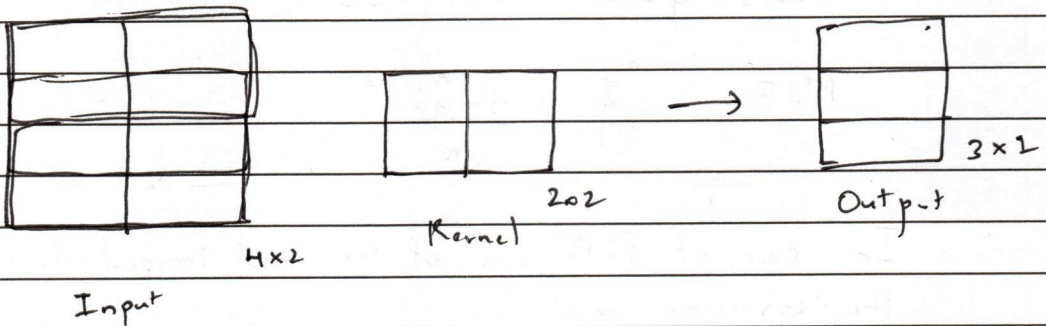
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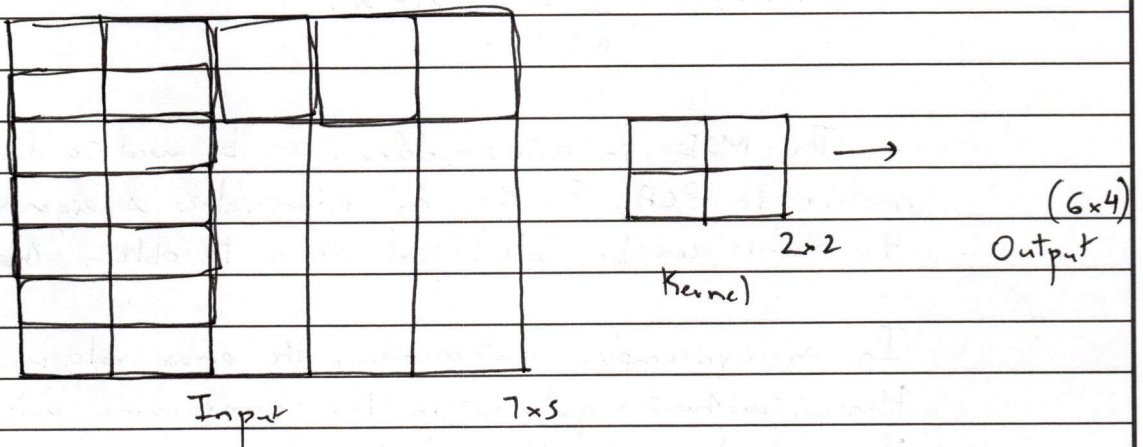
Q.2) a) Input = $H \times W \times C$
Filter size = $F \times F$
No. of filter = K
 $\therefore$ Total filter = $F \times F \times K$.

Let's consider two eg's:-

i) Input = $4 \times 2 \times 3$
Filter = $2 \times 2 \times 10$



ii) Input = $7 \times 5 \times 3$
Filter = $2 \times 2 \times 10$.



$\therefore$ For image $\rightarrow H \times W \times C$
filter $\rightarrow F \times F \times K$

$\therefore$ Output = $[H-F]+1 \times [W-F]+1 \times K$
 $\downarrow$
 floor function

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Q.2)b) Autoencoder becomes equivalent to PCA under the following conditions:

- i) Linear encoder
- ii) Linear decoder
- iii) Error is calculated using mean squared error (MSE)

It means that $g(x)$ & $h(y)$ must be linear functions & the error is:

$$\text{Error} = (y - \hat{y}) \rightarrow \text{MSE}$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

In case of PCA, one of the most important component is the covariance matrix:

$$\text{Cov} \begin{bmatrix} x & y \end{bmatrix} = \begin{bmatrix} \text{Cov}(x, x) & \text{Cov}(x, y) \\ \text{Cov}(y, x) & \text{Cov}(y, y) \end{bmatrix}$$

$$\text{Cov}(x, x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$\therefore$ The MSE in autoencoder can be used to derive the covariance matrix in PCA; Further the eigen values & eigen vectors of this matrix can be calculated so as to obtain the principal component.

In an overcomplete autoencoder, the error rate is minimal. Hence, without regularization the error come out to be 0 & 1 & reach towards the identity matrix.

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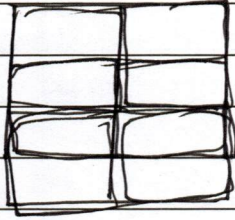
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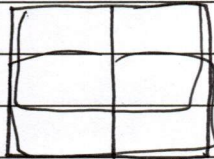


4×2

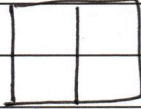


2×2

~~3×3~~
 3×1

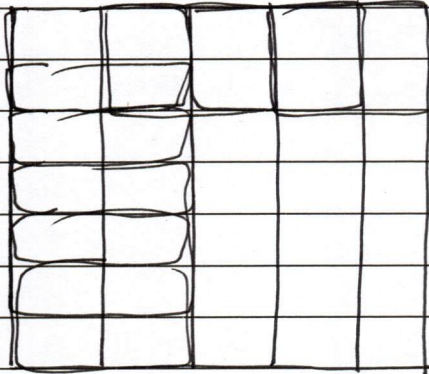


3×2

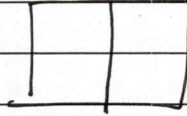


2×2

2×1



7×5



2×2

2×4